



Optima Academy Online
AN EDUCATION EXPERIENCE COMPANY

Biology 1 · Teacher Manual

Florida Course 2000310 · Course Kit 2000310-s2 · **Semester 2 Edition** — Modules 6–10 + Week 36 Final

Teacher copy. Contains answer keys and grading guidance. Do not publish to students.

This document is for the teacher of record only. It is never uploaded to Canvas as student-facing content and never shared with families or students directly.

TABLE OF CONTENTS

- | | |
|--|--|
| 0. How to use this manual & honest limits | 7. Quizzes, discussions & the Final — how to run & grade |
| Adapting to a Live, 4-Day-a-Week Classroom (Modules 6–10 + Week 36) | 8. Coach, not catch |
| 1. How this course works (the Course Kit) | 9. Weekly announcement scripts (all 21 weeks) |
| 2. Ungraded interactives & module wrappers | 10. Teacher workflow & workload map |
| 3. AI & VR in this course (student-facing) | 11. Standards coverage map |
| 4. Using Ignite AI to support students | 12. What makes this course classical |
| 5. Grading categories & gradebook setup | 13. Sensitive-content flags & scripts |
| 6. Rubrics for every teacher-graded item | Appendix. Open items register |

SECTION 0 · HOW TO USE THIS MANUAL

This manual documents the Biology 1 Semester 2 Course Kit as it is built today: the kit specification (2000310-s2, store optimaondemand/optima-courses-2026-27) and the source build files in 03_Build/. Every count, title, point value, and structural claim below was read out of that specification or those files — nothing

here is aspirational. Where something the course design calls for is missing or untested, this manual says so directly and marks it rather than describing the course the design documents imagined. Anything marked **[TO BE CONFIRMED]** is an open item to confirm with the course owner before you rely on it.

Scope of this edition — read this first. Biology 1 is a 36-week course. This manual covers **Semester 2 only**: Modules 6–10 (course Weeks 16–35) and the Week 36 Final — 21 weeks in all. Semester 1 (Modules 1–5 and the Week 15 Midterm) has its own manual and is not repeated here, except where a Semester 2 item reaches back to it (28 Semester 1 questions sit inside the cumulative Final).

Honest limits — stated plainly.

- **The Canvas import test has not been run.** The kit specification has been generated and checked against the build files, but it has not yet been imported into a Canvas course. Treat every description of “what Canvas will show” as the specification’s intent, to be verified on first import.
- **No Canvas course exists for Semester 2 yet.** Unlike the Semester 1 edition (written about a live course), there is no Canvas course 215 equivalent for these 21 weeks; the course reaches you as a Course Kit you import.
- **The 28 Semester 1 questions inside the Final have not been separately QA’d in this build.** The Semester 2 content went through a content QA pass on 2026-10-09; the Final’s Module 1–5 items are reused unchanged from the Semester 1 quizzes.
- The kit carries **no due dates** (none appear in the specification). Due dates in this manual are the intended due *weeks*; you set the calendar dates.

How to read this manual: Sections 1–3 orient you to what a student experiences. Sections 4–9 are operational — the things you configure or do weekly. Section 10 helps you plan your own time across the 21 weeks, with the arithmetic shown. Sections 11–13 are reference material: standards, classical-pedagogy rationale, and content to flag for families in advance. The live-classroom module guides directly after this section carry the per-module teacher guidance (weekly pattern, CER, quarterly project, discussion, misconceptions, framing notes).

This course was built for asynchronous delivery, and the course specification states that live sections use the same curriculum unchanged. The content does not need rewriting for a live room — it needs re-sequencing. Every normal week contains the same ingredients (4 readings, 4 lessons, 4 paired assignments, 1 reading quiz, 2 VR activities), and a live class meeting four days a week redistributes them across class periods instead of a self-paced week. Below is one pattern per module; module-specific notes (CER placement, quarterly project, discussion, misconceptions, and callouts that need your framing) follow each pattern. The pacing suggestions are a plan for the room, not something the kit enforces.

General template (any normal week): **Day 1** — Readings 1–2 and Lessons 1–2; short warm-up discussion. **Day 2** — Readings 3–4 and Lessons 3–4; VR-A (Build/Model) done in class as a hands-on activity. **Day 3** — work time on the 4 paired assignments, VR-B (Explore/Observe), and CER or discussion workshoping in the weeks flagged below. **Day 4** — reading quiz (all 4 lessons), synthesis discussion, preview of next week.

► Module 6 · Evolution and Natural Selection WEEKS 16–18

Weeks 16–18. Week 16 — the fossil record, comparative anatomy, biogeography, molecular evidence (four independent lines). Week 17 — natural selection, the three shapes of selection and artificial selection, drift and gene flow, antibiotic resistance. Week 18 — speciation, Hardy–Weinberg, the origin of life, human evolution.

Weekly pattern. Week 16: Day 1 Lessons 16.01–16.02 (rocks, bones); Day 2 Lessons 16.03–16.04 and VR-A (the four-station “evidence for evolution” build); Day 3 assignments and VR-B (hominid skull sequence); Day 4 quiz. Week 17: Day 1 Lessons 17.01–17.02 (use the Explore interactive in 17.02); Day 2 Lessons 17.03–17.04 and VR-A (peppered-moth scene); Day 3 assignments and VR-B (antibiotic-resistance shift); Day 4 quiz. Week 18: Day 1 Lessons 18.01–18.02 (the 18.02 allele-frequency simulator); Day 2 Lessons 18.03–18.04 (origin of life, human evolution) and VR-A (cladogram of speciation events); Day 3 CER workshop and discussion; Day 4 quiz.

CER. *Three Witnesses, One Story: The Case for Common Descent* — Level 3 open-inquiry, 12 pts, Word via OneDrive. Assigned Week 16 Day 1, due end

of Week 18. Lesson 16.04 is the marked CER anchor. VR-A in Week 16 is built on the same architecture as the CER (claim, three independent pieces of evidence, reasoning) — tell students to keep their notes from it. Workshop Week 18 Day 3: have pairs swap outlines and check that each line of evidence names a specific example.

Discussion D6 — *Which Line of Evidence Persuades You Most?* Due end of Week 18. Post early in the week (Day 1–2) so replies have time; see Section 7.

Common misconceptions to watch. “Survival of the fittest” means the physically strongest (fitness is reproductive success); natural selection creates traits on demand or changes individuals (it acts on existing variation, and populations evolve, not individuals); the antibiotic “taught” bacteria to resist (resistant cells were already present); homologous and analogous structures confused; **“humans evolved from chimpanzees”** (we share a common ancestor roughly 6–7 million years ago, per Lesson 18.04); evolution has a goal or ladder (it is a branching bush); Miller–Urey “created life” (it showed building blocks can form; no cell appeared). Allopatric speciation and the biological species concept are the Week 18 quiz items most likely to expose gaps.

Callouts that need your framing. 16.03 — Darwin and Wallace (“two naturalists, one idea, and the courage to publish”); 16.04 — AI Awareness on machine learning and the tree of life plus a hedged contemporary-research callout on ancient DNA; 17.04 — contemporary research on tracking resistance across the globe; 18.03 — Pasteur’s swan-neck flask and an origin-of-life lesson that is deliberately and explicitly “an honestly open question.” Let students see the line between the well-supported science of how life diversified and the open research on how it began. See Section 13, Flag 1.

► Module 7 · Classification and Diversity of Life

WEEKS 19–22 · Q3 PROJECT

Weeks 19–22. Week 19 — Linnaean hierarchy, binomial nomenclature, using and constructing dichotomous keys. Week 20 — cladograms and phylogenetic trees, monophyletic groups, molecular phylogenetics and clocks, classification as a living science. Week 21 — the three domains,

Bacteria and Archaea, protists, fungi. Week 22 — plant and animal kingdoms, viruses, Florida biodiversity.

Weekly pattern. Week 19: Day 1 Lessons 19.01–19.02; Day 2 Lessons 19.03–19.04 and VR-A (classification ladder); Day 3 key out unknown specimens (VR-B does this virtually with five unknowns; a physical set of leaves or shells is a natural live companion — a suggestion, not a built resource); Day 4 quiz. Week 20: Day 2 VR-A — the cladogram from a character matrix. The correct tree is **(Lamprey, (Salamander, (Lizard, Mouse)))**, ordered by four limbs, amniotic egg, hair; check that students designate the outgroup. Week 21: Day 2 VR-A (three domains side by side) is the molecules-versus-morphology build that anchors the CER. Week 22: Day 1 plants and animals; Day 2 viruses and Florida biodiversity; Day 3 CER workshop plus D7; Day 4 quiz.

CER. *Reading Life by Its Molecules: The Case for Three Domains* — Level 3 open-inquiry, 12 pts. Assigned Week 19 Day 1, due end of Week 22. Anchor content arrives Week 20 (Woese) and Week 21 (three domains); workshop Week 22 Day 3, which is also a heavy week (see Section 10).

Quarterly project Q3 — The Household of Life. Assigned Week 19, **due end of Week 27**, 16 pts, rubric (Section 6). Milestones from the project page: Weeks 19–20 build the classification ladder and cladogram skeleton; Weeks 21–22 three domains plus kingdoms plus dichotomous-key checkpoint (teacher feedback checkpoint); Weeks 23–25 plant anatomy-and-transport station; Week 26 revise, record the guided tour, submit; Week 27 gallery walk and closing reflection. Hold a live gallery walk in Week 27. Note the project spans Module 7 into Module 9, so it overlaps the Module 8 CER and the opening of Module 9.

Discussion D7 — *Is a Virus Alive? Making a Courteous Case*, due end of Week 22.

Common misconceptions to watch. Reversing broad and specific in the hierarchy; writing the species name wrong (capitalized epithet — *Canis lupus*); reading a cladogram as a ladder with one “most advanced” tip, or reading relatedness from tip order instead of nodes; treating the molecular clock as exact; **giving Archaea a nucleus or bracketing Archaea with Bacteria because they look alike**; “all microbes are harmful”; “protists are

one clean group"; "fungi are plants"; "mosses grow tall" and "ferns or mosses flower"; "most animals are vertebrates"; a virus is a cell or a bacterium, and antibiotics work on viruses. The Q3 project page itself warns about treating the three domains as "small/medium/large" and notes the plant station is where most points are won or lost.

Callouts that need your framing. 19.01 and 22.01 — Linnaeus; 20.03 — AI Awareness on algorithms that build trees (a computed tree is a hypothesis, not a fact); 20.04 — Woese and the third domain; 21.02 — Leeuwenhoek, the draper who saw a new world; 21.04 — contemporary research on the "wood-wide web" (hedged: "recent research suggests"; keep it hedged when you retell it); 22.02 — Aristotle's first great sort; 22.04 — AI Awareness on whether an app can name a species. Lesson 22.03 ("Viruses: Are They Alive?") presents a real scientific disagreement; D7 asks students to defend either side.

► Module 8 · Plant Biology

WEEKS 23–25

Weeks 23–25. Week 23 — plant cell features, the three tissue systems, roots and stems, leaf anatomy. Week 24 — xylem and cohesion-tension, transpiration/stomata/the potometer, phloem and pressure-flow, soil and the plant microbiome. Week 25 — flowers and pollination, double fertilization/seed/fruit/germination, hormones, tropisms.

Weekly pattern. Week 23: Day 2 VR-A (leaf cross-section); Day 3 VR-B (root-to-leaf tour). Week 24: Day 1 Lessons 24.01–24.02; Day 2 potometer work (VR-B runs the potometer under varied conditions; a physical potometer demonstration, if you have the equipment, is a good live companion — the Q3 project page recommends a transpiration-pull lab and ties it to the potometer CER); Day 3 workshop the CER data; Day 4 quiz. Week 25: Day 2 VR-A (flower and double fertilization); Day 3 VR-B (phototropism and auxin) plus D8; Day 4 quiz.

CER. *Reading the Potometer: How the Environment Drives a Plant's Water Loss* — Level 3 data analysis, 12 pts. Assigned Week 23 Day 1, due end of Week 25. The class dataset is in the assignment: baseline 4.0 mm/min; wind 8.5; 30°C 6.0; humidity raised 1.5; light removed 1.0. A strong essay compares each to the baseline — wind about $2.1 \times$ ($8.5/4.0$), temperature $1.5 \times$, humidity

about $0.38\times$, darkness $0.25\times$ (my arithmetic from the table, for your checking) — and justifies cohesion-tension.

Discussion D8 — *Can a Plant Sense Without a Brain?*, due end of Week 25.

Common misconceptions to watch. The xylem/phloem reversal (xylem: dead cells, water and minerals, up; phloem: living cells, sugars, source to sink) — the single most common error in this module, and the VR-A note flags it; transpiration as only a problem rather than the driving mechanism; cohesion-tension replaced by “roots pump it up”; the humidity result predicted wrong; palisade versus spongy mesophyll swapped; male/female flower parts reversed (stamen male; carpel/pistil female); pollination confused with fertilization; auxin gathering on the *lit* side (it gathers on the shaded side); plants cannot use N_2 directly (nitrogen-fixing bacteria are required).

Callouts that need your framing. 23.04 — Malpighi and Grew, the first plant anatomists; 24.01 — Stephen Hales; 24.04 — contemporary research on microbes and hardier crops, plus AI Awareness on whether a model can diagnose a sick crop (with an “interrogate it” widget); 25.04 — Darwin, his son, and a 46-year question about plants.

► Module 9 · Human Body Systems

WEEKS 26–29

Weeks 26–29. Week 26 — levels of organization and tissues, homeostasis and feedback, skin, skeletal and muscular systems. Week 27 — digestion and enzymes, circulation, cardiovascular health, respiration and gas exchange. Week 28 — kidney and nephron, neuron and action potential, synapses and how drugs act, endocrine system. Week 29 — innate immunity, adaptive immunity, vaccines, the reproductive system and body-systems integration.

Weekly pattern. Week 26: Day 2 Lessons 26.03–26.04 and VR-A (organization hierarchy); Day 3 VR-B (walk a negative-feedback loop). Week 27: Day 2 VR-A (four-chamber heart walk); Day 3 VR-B (alveolar gas exchange and the digestive tract). Week 28: Day 1 kidney and neuron; Day 2 Lessons 28.03–28.04 and VR-A (build a neuron); Day 3 VR-B (a drug at a synapse), Q4 project kickoff; Day 4 quiz. Week 29: Day 1 innate and adaptive immunity;

Day 2 vaccines and integration with VR-A (vaccine and memory); Day 3 CER workshop plus D9; Day 4 quiz.

CER. *From Synapse to Addiction: Tracing a Drug Through the Brain* — Level 4 contested, the least-scaffolded essay of the module sequence, 12 pts.

Assigned Week 26 Day 1, due end of Week 29. Lesson 28.03 is the marked anchor. **Teacher decision point:** students choose from the four drug groups in the assignment (nicotine; caffeine; cocaine/amphetamines; opioids). Any other drug requires your approval, and the assignment says that approval should depend on the student being able to describe its synaptic action correctly. Keep every approval conversation at the level of mechanism. Plan workshop touchpoints Week 28 Day 3 and Week 29 Day 3. Caffeine is a different chain (adenosine-receptor blocking, more adenosine receptors, tolerance and withdrawal), not a dopamine flood — students who pick it and describe dopamine are making the most likely error.

Quarterly project Q4 — Florida Ecosystem Energy-Flow Walk is assigned Week 28; it is covered under Module 10.

Discussion D9 — *Homeostasis and Self-Government*, due end of Week 29.

The prompt asks students to compare a feedback loop to one conscious habit; the teacher notes ask you to remind students not to share personal medical information.

Common misconceptions to watch. Negative feedback “keeps things exactly constant” (it opposes a change); receptor versus control center confused; the response described as adding to the change; organ placed before tissue, or organ confused with organ system; ligament versus tendon; **arteries always carry oxygenated blood** (pulmonary arteries do not); right and left heart sides mixed; breathing equated with cellular respiration; action potential treated as variable in size (it is all-or-none); signal direction reversed; agonist versus antagonist confused, and a reuptake blocker described as adding a new signal; antibodies credited with memory (memory B and T cells hold it); skipping clonal selection; calling the fast phagocyte stage adaptive; **“a vaccine gives you the disease”**; the nervous system and endocrine system “do the same thing” (fast electrical versus slower chemical).

Callouts that need your framing. 26.02 — the discovery of the steady inside (the course plan names Bernard and Cannon); 27.02 — Harvey counts the blood against 1,400 years of Galen; 27.03 — contemporary research frontier on cardiovascular health; 28.03 — hedged research on rethinking addiction as a brain condition; 28.04 — Banting and Best; 29.02 — Jenner and a milkmaid's hands; 29.03 — the mRNA platform (hedged: "much of this work is still early and unproven") and AI Awareness on drug discovery and disease modeling. See Section 13, Flags 2–4.

► Module 10 · Ecology and Environmental Science

WEEKS 30–35 · Q4

PROJECT

Weeks 30–35. Week 30 — levels of ecological organization, species interactions, Florida ecosystems, biomes and climatographs. Week 31 — food webs, the 10% rule, biomagnification and keystone species, the Yellowstone wolves. Week 32 — carbon, nitrogen, water and phosphorus cycles, human disruption. Week 33 — exponential and logistic growth, carrying capacity, succession, climate-change evidence. Week 34 — biodiversity, causes of loss, Florida invasives, conservation. Week 35 — *Silent Spring*, ecosystem services, the year-spine synthesis, humans as members and shapers.

Weekly pattern. Each week: Day 1 and Day 2 carry two lessons each; VR-A on Day 2; VR-B and assignment time Day 3; quiz Day 4. Specific hooks: Week 30 Day 2 — build a Florida ecosystem (VR-A) and use the biomes Explore interactive in 30.04; Week 31 Day 3 — walk the Yellowstone cascade (VR-B) and begin D10A; Week 32 Day 2 — build a cycle with closed loops (VR-A); Week 33 Day 3 — the climate lesson (33.04) is the CER anchor, so use it as a seminar; Week 35 Day 2 — the year-spine walk (VR-A).

CER. *Three Independent Witnesses: The Case for Human-Driven Climate Change* — Level 4 contested, 12 pts. Assigned Week 30 Day 1, due end of Week 35: the longest runway of the semester. Plan three workshop touchpoints: Week 32 Day 3 (after the carbon cycle), Week 33 Day 3 (after the climate-evidence lesson), Week 35 Day 3. The evidence bank has three lines — the temperature record (about 1.1°C rise since the late 1800s), the CO₂ record and its carbon fingerprint (Keeling Curve begun 1958 at Mauna Loa,

roughly 315 ppm to over 420 ppm; pre-industrial about 280 ppm from ice cores), and shifts in living things (ranges and timing). The assignment states the framing honestly: strong consensus on the evidence; open uncertainty mostly about rate and magnitude; the genuinely open question is what to do, and the essay asks for a reasoned stewardship position. **Grade the reasoning, not the stance.**

Quarterly project Q4 — Florida Ecosystem Energy-Flow Walk. Assigned Week 28, **due end of Week 35** (before the Final; no building during exam week), 16 pts. Four stations: trophic energy pyramid with the 10% rule; a biogeochemical cycle with one human disruption; a human-homeostasis negative-feedback loop; a stewardship station with one Florida conservation issue — plus an “atoms to ecosystems” capstone marker. Milestones: Weeks 28–29 homeostasis station; 30–31 ecosystem and pyramid (teacher feedback checkpoint); 32–34 cycle and stewardship stations; 35 revise, record, submit, gallery walk. The project page warns the most common failure is “a beautiful ecosystem with no flow” — hold the line on the 10% rule and on visible cycle arrows.

Discussions. D10A *The Wolves of Yellowstone: Cascade or Coincidence?* due end of Week 32; D10B *Is the Land Ethic Science, Ethics, or Both?* due end of Week 35.

Week 35 is the heaviest week of the semester. Four things land together: the Module 10 CER, the Q4 capstone project, D10B, and the Week 35 quiz — and the next week is the Final. If you control the calendar, consider asking students to submit the Q4 project at the end of Week 34 or moving the gallery walk to Week 36 Day 1. That is a suggestion, not something the kit specifies.

Common misconceptions to watch. Habitat versus niche; biotic versus abiotic (a dead log with decomposers is treated as biotic); commensalism versus parasitism; predation versus competition; **“energy is recycled” (energy flows one way and is lost as heat; only matter cycles)**; decomposers as consumers; the 10% rule misapplied; the toxin is worst in producers (biomagnification concentrates it at the top); water is “used up”; organisms can use N_2 directly; phosphorus has no gas phase; algae blamed directly for eutrophication oxygen loss (the decomposition of the dead algae

uses it up); the logistic curve drawn as a J-curve; carrying capacity as a fixed wall; primary versus secondary succession (the soil test); fossil-fuel burning “makes new carbon”; “DDT poisoned adults directly” (the Silent Spring mechanism is eggshell thinning and accumulation up the web).

Callouts that need your framing. 30.01 — Humboldt (nature as a web, per the course plan); 31.01 — Elton; 32.01 — Priestley, Lavoisier and Ingenhousz “weighing the air”; 32.02 — Hellriegel and Wilfarth and the legume question; 32.04 — eDNA research (“still maturing”); 33.04 — climate research callout plus AI Awareness on species-range models; 34.03 — AI Awareness on spotting invaders; 34.04 — Leopold’s land ethic; 35.01 — Carson and *Silent Spring*; 35.02 — restoration-ecology research (“promising but far from guaranteed”); 35.04 — the year-end “Tool, Not Oracle” AI reckoning. See Section 13, Flag 5.

► Week 36 · Final Review & Exam FINAL WEEK

There is no new content this week, so the 4-reading/4-lesson pattern does not apply. The kit module “Final Review & Exam · Week 21” contains exactly three items: the review lesson (*The Whole Argument, Atoms to Ecosystems*), the review activity (20 pts), and the Final Exam (88 pts).

Suggested live sequence. Day 1 — deliver the review lesson as a whole-class tracing of the spine, module by module (M1 how we know; M2 atoms and molecules; M3 the cell; M4 energy; M5 information and heredity; M6 evolution; M7 diversity; M8 plants; M9 the human body; M10 ecosystems); start the review activity. Day 2 — finish the review activity in small groups; open floor for the weakest-confidence topics. Day 3 — structured practice from retired reading-quiz items; address the misconception lists in this section directly, especially the Semester 1 anchors (the Final draws 28 items from Modules 1–5). Day 4 — administer the Final (70 MC + 3 CER) in one proctored block. The build specifies no time limit **[TO BE CONFIRMED]**; plan the block for the CER paragraphs of about 150–250 words each.

Review activity. Three parts: A (order the ten module big-ideas, 10 pts), B (name the bridge across five module crossings, 5 pts), C (write the year’s argument in five to seven sentences using at least five listed terms, 5 pts). The

answer key is on the page: bridges $M4 \rightarrow M10 = A$, $M5 \rightarrow M6 = B$, $M6 \rightarrow M7 = C$, $M3 \rightarrow M9 = D$, $M8 \rightarrow M10 = E$. It is a text-entry/upload assignment, so you check it against the key rather than receiving auto-scores. It asks students to complete it before taking the Final.

SECTION 1 · HOW THIS COURSE WORKS

How the kit is structured

The course is delivered as a **Course Kit import** — kit id 2000310-s2 in the optimaondemand/optima-courses-2026-27 store. There is no pre-built Canvas course; you import the kit and Canvas creates the modules, pages, assignments, quizzes and discussions. The structure is:

- **One Canvas module per course week** — 21 weekly modules (20 instructional weeks plus the final week), titled “Module N: <name> · Week K: <topic>” with **semester-relative week numbers**: course Week 16 appears as Week 1, and course Week 36 as Week 21. The final week reads “Final Review & Exam · Week 21.” Lesson, reading and assignment numbers keep **course numbering** (Lesson 16.01 appears in Week 1 of the kit). This offset is the single most likely source of confusion — the table below maps them.
- A 22nd module, **Teacher Resources**, is **unpublished**: the Final Exam Draw Specification, the Final Exam Item Map, and the course syllabus PDF. Keep it unpublished. The specification notes that this teacher manual PDF is bundled there when present **[TO BE CONFIRMED]**.
- Pages (readings, lessons, the Final review lesson) and the content of assignments are **iframes** of optimaondemand.github.io/hs-biology-1-2000310; students submit through the Canvas assignment, not the embedded page. No Module 6–10 module intro/outro wrapper quizzes are in the kit (Section 2).
- Completion requirements are set on module items: pages “must mark done,” assignments and quizzes “must submit,” discussions “must contribute.” Modules are not sequential, so students can move freely.

A typical lesson

Every lesson page (for example Lesson 28.03, “Synapses, Neurotransmitters & How Drugs Act”) follows the same shape: a header naming the quarter, module, lesson, topic and the module’s virtue thread; 2–4 “I will be able to...” objectives; a “Read this first” pointer to the paired Reading page; core content in labeled parts; “Quick check-in” and “Try it / Match it / Check it” self-check widgets with instant feedback (ungraded); history-of-science, contemporary-research and AI-awareness callouts where the content genuinely supports one; a “Your VR This Week” pointer; and a closing “what’s next.” Lessons that anchor a module CER say so in a banner. The graded touchpoint is never on the lesson page itself — it is the separate paired Assignment.

A typical week (what the kit contains)

Item	What the student does	Graded?
4 Reading pages	Reads the Optima-voice adaptation of that lesson’s content	No — anchors the quiz
4 Lesson pages	Works through content, self-check widgets, callouts	No
4 Paired assignments (Lesson N.01–N.04)	Completes one short task per lesson (for example, Assignment 16.01 is about 15–20 minutes; each page carries success criteria and a short rubric)	Yes — 10 pts each, completion / minimal review
1 Reading quiz	9- or 10-item EOC-format MC quiz covering all 4 readings; 1 point per question, 1 attempt ; students see correct answers after	Yes — auto-graded
VR-A (Build/Model)	Constructs a 3D model of the week’s concept; submits a snapshot or short recording	Yes — 10 pts, completion
VR-B (Explore/Observe)	Enters a scene, observes, captures evidence	Yes — 10 pts, completion

Most weeks therefore hold 8 pages, 4 paired assignments, 1 quiz and 2 VR assignments — 15 items. Weeks that open a module also carry the module CER (and the Q3 or Q4 project in Weeks 19 and 28), and the weeks that close a discussion window carry the discussion, so those weeks run to 16–17 items.

Course Week to kit week map

Course week	Kit week	Module	Topic (from the kit module title)	Quiz items
16	Week 1	M6 · Evolution and Natural Selection	Evidence for evolution	9
17	Week 2	M6 · Evolution and Natural Selection	Natural selection & other mechanisms	9
18	Week 3	M6 · Evolution and Natural Selection	Speciation & origins	9
19	Week 4	M7 · Classification and Diversity of Life	Taxonomy & nomenclature	10
20	Week 5	M7 · Classification and Diversity of Life	Phylogenetics	10
21	Week 6	M7 · Classification and Diversity of Life	The three domains & prokaryotic/protist/fungal diversity	10
22	Week 7	M7 · Classification and Diversity of Life	Plant, animal & viral diversity	10
23	Week 8	M8 · Plant Biology	Plant body & tissues	9
24	Week 9	M8 · Plant Biology	Transport & transpiration	9
25	Week 10	M8 · Plant Biology	Plant reproduction & response	9
26	Week 11	M9 · Human Body Systems	Organization, homeostasis & support	10
27	Week 12	M9 · Human Body Systems	Fueling & transporting	10
28	Week 13	M9 · Human Body Systems	Regulation & signaling	10
29	Week 14	M9 · Human Body Systems	Defense & integration	10
30	Week 15	M10 · Ecology and Environmental Science	Levels, interactions, biomes	9
31	Week 16	M10 · Ecology and Environmental Science	Energy flow	9

Course week	Kit week	Module	Topic (from the kit module title)	Quiz items
32	Week 17	M10 · Ecology and Environmental Science	Biogeochemical cycles	9
33	Week 18	M10 · Ecology and Environmental Science	Populations & succession	9
34	Week 19	M10 · Ecology and Environmental Science	Biodiversity & human impact	9
35	Week 20	M10 · Ecology and Environmental Science	Silent Spring & synthesis	9
36	Week 21	Final Review & Exam	Review lesson, review activity (20 pts), Final Exam (88 pts)	73 prompts

Major assessments

Item	Module / due	What it asks	Pts	Teacher time (estimate)
Module 6 CER — Three Witnesses, One Story	M6 · assigned Wk 16, due end Wk 18	Evaluate at least three independent lines of evidence and build one argument for common descent	12	~5 min/student, rubric
Module 7 CER — Reading Life by Its Molecules	M7 · Wk 19 to end Wk 22	Justify the three-domain system: molecular sequence data versus morphology	12	~5 min/student
Module 8 CER — Reading the Potometer	M8 · Wk 23 to end Wk 25	Analyze potometer data; justify cohesion-tension	12	~5 min/student
Module 9 CER — From Synapse to Addiction	M9 · Wk 26 to end Wk 29	Trace a drug from synapse to effect to the basis of addiction; reason about self-government	12	~5 min/student
Module 10 CER — Three Independent Witnesses	M10 · Wk 30 to end Wk 35	Evaluate three independent lines of evidence for human-driven climate change; take a stewardship position	12	~5 min/student
Q3 VR Project — The Household of Life	Assigned Wk 19, due end Wk 27	Three-domain walk plus plant anatomy-and-transport station	16	Rubric (assumed ~3 min)

Item	Module / due	What it asks	Pts	Teacher time (estimate)
Q4 VR Project — Florida Ecosystem Energy-Flow Walk	Assigned Wk 28, due end Wk 35	Energy pyramid, cycle, homeostasis and stewardship stations; atoms-to-ecosystems close	16	Rubric (assumed ~3 min)
6 Discussions (D6, D7, D8, D9, D10A, D10B)	Due end Wk 18, 22, 25, 29, 32, 35	Initial post (about 100–200 words) plus a reply (about 50–100 words)	10 each	~30 s per post
Final Review Activity	Wk 36	Order the spine; name the bridges; write the year's argument	20	~1–2 min check against the key (assumed)
Final Exam	Wk 36	70 MC (auto) + 3 CER short responses (6 pts each)	88	~2 min per CER item

Teacher-time figures follow the Optima tier design (about 5 minutes for a Tier 3 written assessment, about 30 seconds per discussion post) and the Final specification (about 2 minutes per CER short response). They are planning estimates, not measured data; the two figures marked “assumed” are mine, because no source states a time for them. Check them against your own pace in the first cycle.

SECTION 2 · UNGRADED INTERACTIVES & MODULE WRAPPERS

Ungraded interactive activities

Lesson pages carry embedded self-check widgets — “Quick check-in,” “Try it,” “Match it,” “Check it,” flip cards, and a few “Explore” links to vetted external simulations (for example the allele-frequency simulator in Lesson 18.02 and the biomes explorer in Lesson 30.04). They are built on the course’s GitHub Pages site and produce no Canvas submission, so there is nothing to grade. Treat them as formation by practice: a student who skips one will not show up as a missing grade but may show up as a weaker quiz or paired-assignment score. Use downstream performance as the signal rather than chasing completion.

Module intro/outro wrappers

This feature is not in the kit. The Optima Canvas design calls for a module intro/outro pair built as a survey-mode Canvas Quiz (an essential-question response, a confidence rating, a virtue reflection) for cohort-level triage. The Semester 2 kit specification contains no such wrappers: its quizzes are the 20 weekly reading quizzes and the Final Exam. Until wrappers are built, use Canvas Quiz Statistics on each weekly reading quiz as your early-warning signal, and use the six discussions (Section 7) as your read on how students are actually reasoning. If you want the outro-triage workflow, it is an open build item for the course owner **[TO BE CONFIRMED]**.

SECTION 3 · AI & VR IN THIS COURSE (STUDENT-FACING)

AI, as students encounter it

This course does not require students to use any AI tool. AI appears only inside lesson callouts, and only where a genuine biological hook exists. Two kinds recur: plain histories/logistics of how a method works, and honest “opportunity and pitfall” guidance. Semester 2 carries eight AI-awareness callouts, each hedged and each framing AI as a tool to interrogate rather than an authority to trust:

Lesson	Callout	The teacher point
16.04	Machine learning and the tree of life	A computed tree is a hypothesis to test against fossils and anatomy, not an authority
20.03	Algorithms that build trees	Garbage in, garbage out; a confident-looking tree can be wrong
22.04	Can an app name a species for you?	A species-ID app is a lead to verify (links to Florida biodiversity)
24.04	Can a model diagnose a sick crop?	Includes an “Interrogate it — weigh the AI claim” exercise
29.03	AI in drug discovery & disease modeling	Opportunity and complication in drug and vaccine research
33.04	Can a model predict where a species will go?	A projection is not a measurement; bias in training data
34.03	Can a model spot an invader?	Includes an “Interrogate it” exercise

Lesson	Callout	The teacher point
35.04	Tool, not oracle — a year-end reckoning	Opportunities (scale, pattern-finding) and pitfalls (hallucinated facts and fabricated citations, training-data bias, over-reliance that hollows out understanding)

Nothing about grading, assignments or objectives requires a student to touch an AI tool.

VR, as students encounter it

Every instructional week carries two VR assignments, described in each lesson’s “Your VR This Week” pointer: **VR-A (Build/Model)** and **VR-B (Explore/Observe)**, 10 pts each, submitted as a low-friction snapshot or short spatial recording — no essay layered on top. Each VR page states what to build or observe, a success checklist, and an expected time (about 30 minutes in-headset, for the page checked). The two Semester 2 quarterly projects (Q3 and Q4) ask students to combine several weeks of concepts into one larger spatial artifact for a teacher-assembled gallery walk, and are scored on the 16-point project rubric (Section 6). This manual describes VR by what the student does, consistent with Optima’s move away from any single named platform.

Wording note for the course owner. Each of the 40 VR-A/VR-B pages and both quarterly project pages (42 pages) includes a short note block listing the common sticking points for that activity. Those blocks sit on the student-facing page, and their heading uses an older term for the teacher rather than the word “teacher.” They are valuable to you (this manual draws on them), but whether to relabel them “Teacher note” or hide them from students is a decision **[TO BE CONFIRMED]**.

SECTION 4 · USING IGNITE AI TO SUPPORT STUDENTS (TEACHER-FACING)

[TO BE CONFIRMED] before relying on this section. Ignite AI’s exact feature set and availability change with Canvas releases and with what Optima has enabled on your instance. Confirm the current feature list in your Canvas admin settings before promising yourself or a co-teacher a specific button. Everything below is described functionally.

You are one teacher serving a large roster. Ignite AI is the tool that makes that ratio feel personal instead of impossible — used on the right tasks, with you always as the final editor.

Drafting individualized feedback

Have Ignite AI draft feedback on a CER essay or a Final short response using the rubric language in Section 6 as the prompt, then edit for accuracy and voice before posting. The Semester 1 manual estimated this turns a 10-minute per-student CER read into roughly 5–6 minutes without giving up your judgment on the score; treat that as an estimate to test in your first module.

Summarizing quiz analytics

Ask Ignite AI to summarize a reading quiz's item analysis to find the item the cohort missed most, then check that item against the misconception list for that module (the live-adaptation section) before deciding whether to re-teach.

Surfacing students who need a nudge

Use gradebook trends (missed assignments, a declining quiz average, an unposted discussion reply) to flag a student, then have Ignite AI draft a warm, specific check-in that you read, edit, and send yourself.

Generating practice variants

For a student who needs another rep (cladograms, dichotomous keys, 10% rule calculations, feedback-loop tracing), ask for 3–5 new items in the style of the reading-quiz bank, then verify each against the correct answer and the relevant misconception before handing it over.

Communicating with families

Draft a clear, kind progress summary or translate a message, then review it yourself. This is especially useful ahead of the sensitive-content advance notices in Section 13. If you use a drafting tool, never paste student names or personal circumstances into it unless school policy allows it **[TO BE CONFIRMED]**.

Standing rule: AI drafts, the teacher decides. Every use above is coach-not-catch (Section 8): find students who need help early, then reach out. Never use Ignite AI, gradebook analytics, or any tool to surveil, rank, or auto-penalize a student. You read, edit, and own everything before it reaches a student or family. This is also

how the course teaches students to treat AI — as a tool to interrogate, not an authority to trust (the Week 35 reckoning).

SECTION 5 · GRADING CATEGORIES & CANVAS GRADEBOOK SETUP

The kit specifies **five weighted assignment groups** that match the syllabus categories, and marks the course gradebook as weighted. Unlike Semester 1, nothing needs to be created by hand — but **verify all of it after import**, because the import has not been tested **[TO BE CONFIRMED]**.

The five groups and what lands in each

Group (as named in the kit)	Weight	Items in this kit	Group points
CER Writing	30%	5 module CERs (M6–M10), 12 pts each, rubric	60
Unit Assessments	25%	80 paired lesson assignments (10 pts each, 800) + 20 weekly reading quizzes (188 questions at 1 pt = 188) + the Week 36 Final Review Activity (20)	1008
Lab & VR Work	20%	40 VR assignments (VR-A and VR-B, 10 pts each, 400) + the Q3 and Q4 VR projects (16 pts each, 32, rubric)	432
Discussion & Participation	15%	6 discussions (D6, D7, D8, D9, D10A, D10B), 10 pts each (initial post 6 + reply 4), rubric	60
Midterm & Final Exams	10%	The Final Exam only: 70 MC (1 pt each) + 3 CER (6 pts each)	88

An honest note on the Exams group. The group keeps its full-year name, but this kit is the second half of the year and the Midterm belongs to Semester 1. In this kit the “Midterm & Final Exams” group therefore holds **only the Final**, and the Final carries the group’s whole 10%. How the Semester 1 and Semester 2 gradebooks combine into a year grade (if they do) is a school decision, not something the kit specifies **[TO BE CONFIRMED]**.

Notes on placement that differ from the Semester 1 build: all 80 paired assignments sit in *Unit Assessments* (the kit does not move lab-type paired assignments into the lab group); the VR assignments and both quarterly projects sit in *Lab & VR Work*; reading quizzes are in *Unit Assessments*; and discussions exist, so

Discussion & Participation is earnable — the empty-15% problem flagged in Semester 1 does not recur here, **provided the six discussions import correctly**.

What one item is worth in the final grade

Canvas weights each group by percentage earned in that group, so an item's share of the course grade is (item points ÷ group points) × group weight. Arithmetic from the counts above:

Item	Calculation	Share of course grade
One module CER (12 pts)	$12 \div 60 \times 30\%$	6.00% (five CERs = 30%)
One discussion (10 pts)	$10 \div 60 \times 15\%$	2.50% (six = 15%)
One VR assignment (10 pts)	$10 \div 432 \times 20\%$	0.46%
One quarterly VR project (16 pts)	$16 \div 432 \times 20\%$	0.74%
One paired assignment (10 pts)	$10 \div 1,008 \times 25\%$	0.25%
One reading-quiz question (1 pt)	$1 \div 1,008 \times 25\%$	0.025% (a 9- or 10-item quiz $\approx$ 0.22–0.25%)
Final Review Activity (20 pts)	$20 \div 1,008 \times 25\%$	0.50%
The Final Exam (88 pts)	whole group	10.00% (each MC point 0.11%; each CER item 0.68%)

Read this plainly: a single missed paired assignment costs a quarter of one percent, while a single missed CER or discussion costs six or two and a half percent. Write your late-work and re-engagement priorities accordingly (Section 8).

Setup steps after import

- **Confirm the groups and weights** — Assignments should show the five groups above and weights summing to 100% with “weight final grade based on assignment groups” on.
- **Set due dates.** The kit carries none. Intended due weeks are in the table below; set calendar dates from your school calendar.
- **Confirm rubrics are attached** — eight rubrics are specified: five module CER rubrics, the Q3 and Q4 project rubrics, and the Discussion Rubric. The Final's

three CER items have *no* Canvas rubric attached in the specification; use Section 6 (or build one).

- **Confirm discussion settings** — threaded, graded, 10 pts, “require initial post before seeing replies.”
- **VR assignments** are specified as 10-point assignments with snapshot/recording/URL submission. Treat them as completion items: a genuine attempt that shows engagement with the target concept earns full points; do not grade production quality.
- **Keep Teacher Resources unpublished.** It contains the Final’s draw specification and item map.
- **Use the Modules progress view** as a weekly scan (page completion, submissions, discussion contributions are all tracked as requirements), not as a grading tool.

Item	Intended due
Module 6 CER · Discussion D6	End of course Week 18 (kit Week 3)
Module 7 CER · Discussion D7	End of course Week 22 (kit Week 7)
Module 8 CER · Discussion D8	End of course Week 25 (kit Week 10)
Module 9 CER · Discussion D9	End of course Week 29 (kit Week 14)
Discussion D10A	End of course Week 32 (kit Week 17)
Q3 VR Project	End of course Week 27 (kit Week 12)
Module 10 CER · Discussion D10B · Q4 VR Project	End of course Week 35 (kit Week 20)
Final Review Activity · Final Exam	Course Week 36 (kit Week 21); activity before the exam
Weekly reading quiz, paired assignments, VR-A/VR-B	Set by you per week; all are in that week’s module

SECTION 6 · RUBRICS FOR EVERY TEACHER-GRADED ITEM

Eight native Canvas rubrics are specified in the kit and attached to their items: five module CER rubrics, the Q3 and Q4 project rubrics, and one Discussion Rubric. The

Final's CER items are scored with the compressed 6-point guide below. All of these are your responsibility to confirm in SpeedGrader after the first import **[TO BE CONFIRMED]**.

The standard module CER rubric (same structure across all five modules)

Criterion	Strong (3)	Developing (2)	Beginning (1)
Claim	Specific, arguable, directly addresses the question	Somewhat specific but vague or incomplete	Missing or merely restates the question
Evidence	Specific, accurate, each piece explained (the number and kind required is module-specific, below)	Partial, or general, or thin	One piece, generic, or inaccurate
Reasoning	Explains the scientific principle (module-specific, below) and handles a counter-consideration where the module asks for one	Links evidence to claim with gaps	Restates evidence without a logical bridge
Integration	One coherent argument, tied to earlier modules, within about 400–700 words	Components present but loosely connected	Listed independently
Total possible	12 pts		

Read once for claim and evidence, once for reasoning and integration; about 5 minutes per student. The criteria and point values stay fixed across modules so grading stays consistent, but each module's "Strong" descriptors are specific. They are reproduced below exactly as specified in the kit (the Developing and Beginning descriptors are on the rubric in Canvas and on each CER page):

Module	Claim (Strong)	Evidence (Strong)	Reasoning (Strong)	Integration (Strong)
M6 · Level 3 open-inquiry	Specific and arguable; asserts common descent supported by converging evidence.	Three or more independent lines, each with a specific named example.	Explains why each line supports common descent AND why convergence of independent lines is powerful; handles homologous vs. analogous.	Reads as one coherent argument and ties to variation (M5) and/or classification (M7); within the word range.

Module	Claim (Strong)	Evidence (Strong)	Reasoning (Strong)	Integration (Strong)
M7 · Level 3 open- inquiry	Specific and arguable; asserts the molecularly based three-domain system over morphology alone.	Contrasts morphology vs. molecular (rRNA) evidence with a specific molecular difference cited.	Explains why sequences record ancestry more reliably than appearance AND fairly weighs morphology's value and limits.	Reads as one coherent argument and ties to common descent (M6) and/or the molecules of M2/M5; within the word range.
M8 · Level 3 data analysis	Specific and data-grounded; names the full pattern (rises with wind/heat/light, falls with humidity/dark).	Cites 3+ specific values, each compared to baseline.	Explains the mechanism for the conditions AND justifies cohesion-tension (pull + cohesion lifting the xylem column).	Reads as one coherent argument and ties to water's properties (M2) and/or the stomata-photosynthesis trade-off (M4); within the word range.
M9 · Level 4 contested	Specific and defensible; names the drug and states what addiction to it fundamentally is.	Synaptic action named correctly (neurotransmitter, receptor, mechanism) plus effect and repeated-use change.	Explains the full causal chain synapse → effect → neuroadaptation → tolerance/dependence.	One coherent argument; ties addiction to homeostasis/self-government and reasons through the contested question; within word range.
M10 · Level 4 contested	Specific and defensible; states what the combined evidence best supports.	Three independent lines, cited concretely with real records/numbers.	Explains mechanisms, argues why convergence is powerful, and fairly weighs a counter-consideration.	One coherent argument; reasons from the biology to a defensible stewardship position; within word range.

Scaffolding ramps from Level 3 (M6–M8) to Level 4 (M9–M10), consistent with the year's guided-to-contested design. The two Level 4 modules ask for a reasoned position on a question beyond the biology (self-government and responsibility in

M9; stewardship in M10). **Grade the quality of reasoning from evidence, not the position taken.**

Quarterly VR project rubric (Q3 and Q4): 4 criteria × 4 levels = 16 pts

Criterion	4 — Exemplary	3 — Proficient	2 — Developing	1 — Beginning
Completeness	All required elements present and clearly labeled (Q3: three domains + kingdoms + plant station + key checkpoint; Q4: pyramid, cycle, homeostasis, stewardship stations + capstone marker)	Most present and labeled	Several missing or unlabeled	Few present
Accuracy (project-specific)	Q3: organisms in the correct domain/kingdom; plant tissues and transport correct. Q4: trophic levels and 10% energy loss correct; cycle flows accurate; homeostasis station shows a valid feedback loop	Minor errors; core relationships correct	Several errors	Major conceptual errors
Iterative building	Recording/snapshots show deliberate construction and revision	Shows the construction process	Limited process shown	Final artifact only
Communication	Tour explains ordering and structure-to-function (Q3), or flows and the organism-ecosystem link, closing the year spine (Q4)	Mostly clear	Partial or unclear	Little or none

High-school standard is recognizable accuracy, not perfection. Submission is a guided recording (about 3–4 minutes for Q3; 4–5 for Q4), two or three named snapshots, a one-line gallery caption, and a short typed reflection (about 150 words for Q3; about 200 for Q4). Score and then assemble the gallery walk; celebrate strong builds publicly.

Discussion Rubric (D6–D10B): 10 pts

Part	Substantive	Thin	Missing
Initial post: claim, specific evidence, reasoning (6 pts)	6 — clear claim; specific evidence from the module; reasoning explains why the evidence supports the claim	3 — claim present but evidence general or reasoning missing/incomplete	0 — no post, or does not address the prompt
Reply to a classmate (4 pts)	4 — responds to a specific point and adds reasoning	2 — generic agreement/disagreement, little reasoning	0 — no reply

Free-response answer guide — the 3 Final CER short responses (6 pts each, 18 total)

Component	0	1	2	3	Max
Claim	No or incorrect claim	Clear, correct, specific; answers the prompt	—	—	1
Evidence	None or irrelevant	Some correct evidence, with gaps	Accurate, specific, names the relevant steps/facts	—	2
Reasoning (mechanism)	None	States <i>what</i> happens but not <i>why</i>	Explains the mechanism with one or more gaps	Explains <i>why</i> each step forces the next; mechanism complete	3

Each response is a focused paragraph of about 150–250 words written during the exam, scored Claim 1 / Evidence 2 / Reasoning 3. The specification puts the load at about 2 minutes per item. What a full-credit response contains (derived from the prompts and the course content; use as a checklist, not a script):

- CER 1 — Following a carbon atom.** Claim: carbon cycles between air and living things while energy flows through once. Evidence: CO₂ is taken in and fixed into glucose by photosynthesis (the Calvin cycle in the chloroplast stroma); glucose is broken down in cellular respiration (glycolysis in the cytoplasm; the citric-acid cycle in the mitochondrial matrix, where CO₂ is released); carbon returns to the air as CO₂. Reasoning: light energy is captured and converted to chemical energy; much of it ends up as heat when cells use it, and that energy is not

recycled, but the carbon atom is reusable matter — “energy flows one way but matter cycles.”

- **CER 2 — From a mutation to common descent.** Claim: accumulated heritable DNA changes make species differ in ways that reveal shared ancestry. Evidence: a mutation arises; it matters over generations only if it is heritable (passed on through meiosis/reproduction); natural selection acts on the variation that exists; differences accumulate in separated lineages; similar DNA sequences and anatomy are what we later read as common descent. Reasoning: why variation must be *heritable* to matter; why the more alike two sequences are, the more recently the species shared an ancestor.
- **CER 3 — Why so few hawks? (the Form used in the kit: energy flow).** Claim: each step in the food chain supports less life because energy is lost. Evidence: grass → grasshopper → mouse → hawk; only about 10% of the energy passes from one trophic level to the next (the 10% rule). Reasoning: the other roughly 90% is used by the organism or lost as heat (and not all of an organism is eaten), so each higher level has less energy to support fewer individuals; the energy first entered through plant photosynthesis, flows one way and leaves as heat, while matter is cycled by decomposers.

SECTION 7 · QUIZZES, DISCUSSIONS & THE FINAL — HOW TO RUN & GRADE

Weekly reading quizzes: 20 quizzes, 188 questions

Every question is EOC-format multiple choice, stands alone as a bank item, and targets a documented misconception. In the kit each question is worth 1 point, quizzes allow **1 attempt**, answers shuffle, and correct answers are shown after submission. (The answer rationales are *not* in the Canvas quiz; they live in each week’s reading-quiz.md in the build folder, which is also your answer key and re-teaching source.)

Course wk	Kit wk	Mod	Quiz topic	Items
16	Wk 1	M6	Evidence for evolution	9
17	Wk 2	M6	Natural selection & other mechanisms	9
18	Wk 3	M6	Speciation & origins	9

Course wk	Kit wk	Mod	Quiz topic	Items
19	Wk 4	M7	Taxonomy & nomenclature	10
20	Wk 5	M7	Phylogenetics	10
21	Wk 6	M7	The three domains & prokaryotic/protist/fungal diversity	10
22	Wk 7	M7	Plant, animal & viral diversity	10
23	Wk 8	M8	Plant body & tissues	9
24	Wk 9	M8	Transport & transpiration	9
25	Wk 10	M8	Plant reproduction & response	9
26	Wk 11	M9	Organization, homeostasis & support	10
27	Wk 12	M9	Fueling & transporting	10
28	Wk 13	M9	Regulation & signaling	10
29	Wk 14	M9	Defense & integration	10
30	Wk 15	M10	Levels, interactions, biomes	9
31	Wk 16	M10	Energy flow	9
32	Wk 17	M10	Biogeochemical cycles	9
33	Wk 18	M10	Populations & succession	9
34	Wk 19	M10	Biodiversity & human impact	9
35	Wk 20	M10	Silent Spring & synthesis	9

Module banks: M6 27 items, M7 40, M8 27, M9 40, M10 54 — 188 for Semester 2; with Semester 1's 126, the full-year bank is 314. Weekly habit: skim Canvas Quiz Statistics for the most-missed item; if it is one of the documented misconceptions in Section "Adapting," spend five minutes re-teaching it rather than moving on.

Paired assignments and VR

Paired assignments (10 pts) are short, objective-aligned tasks; each page lists success criteria, which in the page checked state the expected answers (for example Assignment 16.01: Layer 4 oldest, Layer 1 youngest, deeper layers are older by superposition; index fossil → B, transitional fossil → A). Spot-check a sample rather than reading every submission line by line. VR-A and VR-B are completion items: confirm the snapshot or recording shows genuine engagement with the target

concept. Use the sticking points in the live-adaptation guides as a quick checklist (for example Week 20 VR-A: the correct tree is (Lamprey, (Salamander, (Lizard, Mouse)))).

Discussions — six native Canvas discussions, 10 pts each

Discussions are the course’s Student-and-Community layer. Each is a threaded, graded discussion in *Discussion & Participation*, with the Discussion Rubric (Section 6): initial post (about 100–200 words; claim, evidence, reasoning) worth 6 and a reply to a classmate (about 50–100 words) worth 4. Students must post before they can see replies. Each prompt asks students to post early in the module’s final week so classmates have time to reply. **Grading guidance:** substantive = full credit; thin = partial; missing = zero. Read in batches, scoring as you read, at about 30 seconds per post. Post **two or three brief public comments** per discussion that model good thinking, correct a misconception, or highlight an excellent contribution — these do triple duty as feedback, culture, and visible teacher presence.

Discussion	Prompt in brief	What to look for · public-comment idea
D6 Which Line of Evidence Persuades You Most? (Courage) Due: end of Week 18	Pick one of the four lines (fossils, anatomy, biogeography, molecules); give a specific example; say what rival explanation it rules out. Reply to a classmate who chose a different line: does independence strengthen the case?	A committed claim; a named specific example (not “fossils show change”); reasoning that excludes an alternative. Thin posts restate the module summary. <i>Public comment:</i> Highlight a post that explains why independence of lines strengthens an argument (the CER connection).
D7 Is a Virus Alive? Making a Courteous Case (Courtesy) Due: end of Week 22	State whether viruses are living; use at least two facts from the reading; say which characteristics of life you treat as essential. Reply to a classmate who disagrees: restate their strongest point fairly first.	No single right answer; grade the reasoning. Watch for factual slips (a virus is not a cell; it replicates only inside a host cell). <i>Public comment:</i> Model the “restate before you respond” move.

Discussion	Prompt in brief	What to look for · public-comment idea
D8 Can a Plant Sense Without a Brain? (Wonder) Due: end of Week 25	Define "sense" in a sentence; walk one plant response from signal to response (tropism, guard cells, or a hormone effect); test it against your definition. Reply to a classmate with a different definition.	Definition first, then testing the example. Confirm the biology: auxin redistribution and shaded-side elongation for phototropism; guard-cell turgor changes for stomata; response without nerves. <i>Public comment:</i> Praise a post that tests its example against its own definition.
D9 Homeostasis and Self-Government (Self-government) Due: end of Week 29	Choose one negative-feedback loop; say what is detected, what responds, how the response reverses the change; explain what happens when it is overwhelmed; compare with one conscious habit (where it holds, where it breaks down).	The response must <i>reduce</i> the original change. Remind students not to share personal medical information; use the module examples. <i>Public comment:</i> Point out a thoughtful place where the habit analogy breaks down (the body's loop works without choice).
D10A The Wolves of Yellowstone: Cascade or Coincidence? (Responsibility) Due: end of Week 32	Rate how strongly the evidence supports a wolf-caused trophic cascade (strongly/partly/weakly); cite at least two effects and one other factor; say what further evidence would help.	The lesson is deliberately hedged: the wolf's keystone role is well supported; the size of downstream effects is still debated (climate, drought, other predators, elk numbers). Strong posts keep those apart. <i>Public comment:</i> Highlight a post that names what evidence would change its mind.

Discussion	Prompt in brief	What to look for · public-comment idea
D10B Is the Land Ethic Science, Ethics, or Both? (Responsibility) Due: end of Week 35	Leopold: "A thing is right when it tends to preserve the integrity, stability, and beauty of the biotic community..." Is that scientific, ethical, or both? Use a module example; explain the gap between what science shows and what people decide. Reply: ask a question that tests a classmate's reasoning, then answer it fairly yourself.	The strongest posts separate descriptive claims (what is) from prescriptive ones (what ought to be) and still connect them with evidence. Keep it on the course's evidence; not about any party or policy. Step in with a public comment if it drifts. <i>Public comment:</i> Close the Responsibility thread; praise a clean is/ought distinction.

Prompt text is in 03_Build/module-XX-*/discussion.md (Modules 6–10); the Module 10 file holds both D10A and D10B.

The Final Exam — how it was assembled, how to score it, how to swap items

Format. 73 prompts, 88 points: **70 multiple choice** (1 pt each, auto-graded) plus **3 CER short responses** (6 pts each, teacher-scored). One attempt; shuffled; correct answers are *not* shown after submission. It counts as the Midterm & Final Exams group (10%).

How the 70 MC were assembled. By a stratified draw from the full-year bank of 314 reading-quiz items (126 Semester 1 + 188 Semester 2), random seed 2000310, per the Final specification (semester-2-final/final-quiz-spec.md). The quota by module was nudged from size-proportional toward the EOC domain weights:

Module	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	Total
MC drawn	3	5	6	6	8	8	7	5	10	12	70

Semester 2 modules supply 42 of the 70 items (8 + 7 + 5 + 10 + 12); Modules 1–5 supply the other **28**. The draw guarantees one "misconception anchor" per module a seat (for example: M6 fitness means reproductive success, selection acts on existing variation, antibiotic resistance as evolution; M7 cladograms group by shared derived characters, the three domains rest on molecular data; M8 cohesion-tension, guard cells, tropisms without nerves; M9 negative feedback, the Na⁺/K⁺ pump in the neuron, antibody specificity, a vaccine does not cause the disease;

M10 energy flows while matter cycles, the 10% rule, carbon links photosynthesis and respiration to the atmosphere, climate evidence from independent lines). Resulting domain shares: molecular/cellular about 36%, classification/evolution about 21%, ecology about 17%, human anatomy about 14%; the last two sit slightly below their EOC bands, as the specification itself states. Two items were deliberately excluded from the draw because they follow a carbon atom and would give away CER 1 (Week 10 Q9 and Week 35 Q8).

Where the answer key is. (1) 03_Build/semester-2-final/final-quiz.md — every MC item with the correct option flagged and labeled with its source week and question number (“Wk 17 Q2”); (2) the kit’s Final quiz carries the same items with correct answers flagged; (3) the unpublished *Teacher Resources* module holds the Final Exam Draw Specification and the Final Exam Item Map (source, module, and anchor/random selection for each of the 70 items); (4) the three CER guides are in Section 6 above and in the specification §5.

CER 3 uses the energy-flow form (“Why So Few Hawks?”). The specification offers two forms: A (three independent lines of evidence for human-driven climate change) and B (energy flow and the 10% rule). The kit uses **Form B** because the MC item from Week 33 Q9 lists the three climate-evidence lines, which would give away Form A. Form B also reuses the Module 8-to-10 and Module 4-to-10 bridges from the review lesson. Alternate forms across sections is allowed by the specification.

How to swap items. Replace an item with another from the same module and, where possible, the same week’s pool so the per-week quotas in the specification (§2a) still hold. Never remove or replace a pinned misconception anchor (they are labeled in the item map). Keep every item distinct within the exam, keep the stem and four options verbatim (the distractors are misconception-based — do not edit), and do not reintroduce the two excluded items. If you swap CER 3 to Form A, replace MC Week 33 Q9 as well. Because the Final may reuse Midterm items, if you want no overlap, exclude the 50 Midterm items from the Module 1–5 pools before swapping (the specification notes this option).

QA status of the Final, plainly. The 42 Semester 2 items were drawn after the 2026-10-09 content QA of Modules 6–10. The **28 Semester 1 items have not been separately QA’d in this build**; they are the vetted Semester 1 quiz items used unchanged. Before the exam, read those 28 items yourself (they are marked “M1”

through "M5" in `final-quiz.md`). The Canvas import has not been run, so also open the imported quiz once and confirm 70 questions, 3 essays, 88 points, 1 attempt, and no correct-answer display.

Per-instance table — the five module CERs and the Final

Module	Item	Scaffolding level	Rubric
M6	Three Witnesses, One Story: The Case for Common Descent	Level 3 (open inquiry)	Module 6 CER Rubric (12 pts)
M7	Reading Life by Its Molecules: The Case for Three Domains	Level 3 (open inquiry)	Module 7 CER Rubric (12 pts)
M8	Reading the Potometer: How the Environment Drives a Plant's Water Loss	Level 3 (data analysis)	Module 8 CER Rubric (12 pts)
M9	From Synapse to Addiction: Tracing a Drug Through the Brain	Level 4 (contested)	Module 9 CER Rubric (12 pts)
M10	Three Independent Witnesses: The Case for Human-Driven Climate Change	Level 4 (contested)	Module 10 CER Rubric (12 pts)
Final	3 short responses (carbon atom; mutation to common descent; why so few hawks)	Timed, compressed CER	Section 6 answer guide (6 pts each)

The scaffolding ramp is non-decreasing across the semester (3, 3, 3, 4, 4), continuing the year's movement toward the contested-question format.

SECTION 8 · COACH, NOT CATCH

Your job is to move each student forward, not to detect and punish. In practice:

- Feedback names the next step, not just the error. Lead with what is working before what needs revision.
- Allow re-attempts on paired assignments — they are a learning tool, not a one-shot trap. (The weekly reading quizzes are specified as one attempt; if a student has a technical problem, a second attempt is your call.)
- Use quiz analytics and Ignite AI to find students who need help *early*, then reach out — never to build a case against them. A student who has stopped posting in discussions is a quieter signal than a missing quiz; notice it.

- On suspected academic dishonesty or AI misuse: open a conversation first. Ask the student to talk through their thinking — for a CER, to explain their evidence in their own words. Treat it as a formation moment, not an accusation-first workflow.
- When you highlight strong public work in a discussion, model good thinking — do not single out weak work for public criticism. Your two or three public comments per discussion are the place to do this well.
- In the contested modules (M9, M10) and in discussions D7, D10A and D10B, protect the room: courteous disagreement is the skill being practiced, and no one is graded for which side they land on.

SECTION 9 · WEEKLY ANNOUNCEMENT SCRIPTS

Reusable template

"Welcome to [Week N · Module Title]. This week you'll work through [brief topic]. The one thing not to miss: [due date/assessment]. [One sentence connecting to a tricky concept or the module's virtue]. [One warm, human line of encouragement]. Questions? [contact method]."

Use **course week** numbers in announcements if your students know the course as a 36-week year, and mention the kit week only if students see "Week 1" in Canvas and ask. Each script below is built from that week's actual topics and due items; fill in dates and your contact.

Drafted scripts — one per week (21)

WEEK 16 · KIT WEEK 1 · MODULE 6 OPENING · COURAGE

"Welcome to Semester 2 and Module 6, Evolution and Natural Selection. This week we ask how we know species share ancestors — and the answer comes from four independent lines of evidence: fossils, anatomy, geography and molecules. Your Module 6 CER, Three Witnesses, One Story, is assigned today and due at the end of Week 18; keep your notes from the Week 16 VR build, because it is the same argument in 3D. Reading quiz due [date]. This module's virtue is courage: following evidence where it leads, as Darwin did for twenty years. Questions any time at [contact]."

WEEK 17 · KIT WEEK 2

"This week: how natural selection actually works — and the common slip to avoid. 'Survival of the fittest' does not mean the strongest; fitness is about reproductive success, and populations evolve, not individuals. You will also watch evolution happen in real time with antibiotic resistance. Quiz due [date]. Your CER draft can start taking shape: which three lines of evidence will you argue from?"

WEEK 18 · KIT WEEK 3 · MODULE 6 CER + D6 DUE

"Big week: your Module 6 CER and Discussion 6, Which Line of Evidence Persuades You Most?, are both due at the end of the week ([due date]). In the CER, name specific examples — a named fossil, a named structure — and make the move that matters most: explain why independent lines converging is stronger than any one alone. Please post your discussion initial post early so classmates can reply. This week also tackles the hardest question honestly — how life began — and where the science is still open. Proud of the courage you have shown."

WEEK 19 · KIT WEEK 4 · MODULE 7 OPENING · COURTESY (INCLUDES Q3 PROJECT)

"Welcome to Module 7, Classification and Diversity of Life. This week: how scientists name and sort the household of life — the Linnaean ranks, two-part scientific names, and dichotomous keys you will use and build. Two big things open today: your Module 7 CER (due end of Week 22) and your Quarter 3 VR project, The Household of Life (due end of Week 27). Give the project steady time rather than a last-week sprint. This module's virtue is courtesy: giving each living thing its proper name and place. Quiz due [date]."

WEEK 20 · KIT WEEK 5

"Trees of relationship this week: cladograms, shared derived characters, and the outgroup. The tip to remember: relatedness is read from the branching points, not from where the tips sit — rotating a branch does not change a tree. You will also meet molecular clocks and the story of how DNA sequences started to redraw the tree of life. Quiz due [date]."

WEEK 21 · KIT WEEK 6

"The three domains: Bacteria, Archaea and Eukarya. Watch for the mix-up that trips up most students — Archaea look like Bacteria but are not their close kin, and they are prokaryotes with no nucleus. You will also meet protists and fungi (and learn why fungi are not plants). Your Week 21 VR build is the molecules-versus-morphology idea that anchors your Module 7 CER. Quiz due [date]."

WEEK 22 · KIT WEEK 7 · MODULE 7 CER + D7 DUE

"Your Module 7 CER, Reading Life by Its Molecules, and Discussion 7, Is a Virus Alive? Making a Courteous Case, are due at the end of this week ([due date]). In the discussion, either answer is fine if you defend it — and in your reply, restate your classmate's strongest point fairly before you respond. In the CER, be fair to morphology and still explain why molecules win when they disagree. This week also covers plants, animals, viruses and Florida's own biodiversity. Quiz due [date]."

WEEK 23 · KIT WEEK 8 · MODULE 8 OPENING · WONDER

"Welcome to Module 8, Plant Biology. This week: what a plant is built from — cell features, three tissue systems, roots, stems, and the layered leaf. Your Module 8 CER is a data analysis: you will read potometer numbers and explain how the environment changes a plant's water loss. It is assigned today and due at the end of Week 25. The virtue is wonder; a tree lifts water dozens of meters with no pump. Quiz due [date]."

WEEK 24 · KIT WEEK 9

"How does water climb a tree? This week: xylem, transpiration, stomata and the potometer, then phloem. One thing to keep straight all week: xylem carries water up; phloem carries sugar from source to sink — the reversal is the most common error in this module. The numbers in your CER start to make sense after Lesson 24.02. Quiz due [date]."

WEEK 25 · KIT WEEK 10 · MODULE 8 CER + D8 DUE

"Your Module 8 CER, Reading the Potometer, and Discussion 8, Can a Plant Sense Without a Brain?, are due at the end of the week ([due date]). In the CER, cite at least three numbers and compare each to the 4.0 baseline. In the discussion, define 'sense' first, then test your example against it. This week is flowers, seeds, hormones and tropisms — and the story of Darwin, his son, and a 46-year question about how plants sense light. Quiz due [date]."

WEEK 26 · KIT WEEK 11 · MODULE 9 OPENING · SELF-GOVERNMENT

"Welcome to Module 9, Human Body Systems. We begin with how your body governs itself: levels of organization, homeostasis, and negative feedback, then skin, bones and muscle. Your Module 9 CER, From Synapse to Addiction, is assigned today and due at the end of Week 29; it asks you to reason about the biology of a drug's action, as science. Your Quarter 3 VR project is still in progress and is due at the end of next week. The virtue is self-government. Quiz due [date]."

WEEK 27 · KIT WEEK 12 · Q3 PROJECT DUE

"Your Quarter 3 VR project, The Household of Life, is due at the end of this week ([due date]). Record your guided tour, take your snapshots and write your short reflection — where did molecular data disagree with appearances? We will walk the gallery together. This week in Module 9: digestion, the heart and circulation, cardiovascular health, and breathing. Remember: pulmonary arteries carry oxygen-poor blood. Quiz due [date]."

WEEK 28 · KIT WEEK 13 · Q4 PROJECT OPENS

"This week: the kidney, the neuron, the synapse, and the endocrine system. Lesson 28.03 is the anchor for your Module 9 CER — read it closely. We study drug action strictly as biology: how a molecule changes a signal, never how any substance is used. Your Quarter 4 VR project, the Florida Ecosystem Energy-Flow Walk, opens today and is due at the end of Week 35; start with the homeostasis station. Quiz due [date]."

WEEK 29 · KIT WEEK 14 · MODULE 9 CER + D9 DUE

"Your Module 9 CER and Discussion 9, Homeostasis and Self-Government, are due at the end of the week ([due date]). For the CER, name your drug and trace the full chain: synapse, effect, the brain's own adaptation, tolerance and dependence — and reason honestly about what understanding the mechanism means for self-government. For the discussion, please share no personal medical information; stay with the module's examples. This week: innate and adaptive immunity, vaccines, and the body as one community. Quiz due [date]."

WEEK 30 · KIT WEEK 15 · MODULE 10 OPENING · RESPONSIBILITY

"Welcome to Module 10, Ecology and Environmental Science — the last stretch of the year's one long argument, from atoms to ecosystems. This week: levels of ecological organization, how species live together, Florida's ecosystems and the biomes of the world. Your Module 10 CER, Three Independent Witnesses, is assigned today and due at the end of Week 35 — a long runway on purpose. Keep your notes as you go. Quiz due [date]."

WEEK 31 · KIT WEEK 16

"Energy flow: who eats whom, the 10% rule, and why food chains are short. The one idea to hold onto: energy flows one way and is lost as heat; matter cycles. You will also meet biomagnification, keystone species, and the wolves of Yellowstone — the case study for Discussion 10A, due at the end of next week. Quiz due [date]."

WEEK 32 · KIT WEEK 17 · D10A DUE

"Discussion 10A, The Wolves of Yellowstone: Cascade or Coincidence?, is due at the end of the week ([due date]). Keep the well-supported part (the wolf's keystone role) apart from the debated part (how big the downstream effects were). This week: the carbon, nitrogen, water and phosphorus cycles — and what happens when humans push them. Remember: plants cannot use nitrogen gas directly. Quiz due [date]."

WEEK 33 · KIT WEEK 18

"This week: how populations grow and stop growing, succession, and the evidence on climate. Lesson 33.04 is the anchor for your Module 10 CER — three independent lines of evidence, and why their agreement matters. We hold the evidence the way this course always does: confident where it is strong, careful where it is open. Quiz due [date]."

WEEK 34 · KIT WEEK 19

"Biodiversity and what threatens it, Florida's invaders (pythons, lionfish and others), and conservation, with Aldo Leopold's land ethic. Your Q4 VR project, Module 10 CER and Discussion 10B are all due at the end of next week, so use this week to get ahead. Quiz due [date]."

WEEK 35 · KIT WEEK 20 · CER + D10B + Q4 PROJECT DUE

"The last regular week of the course. Three things are due at the end of it ([due date]): your Module 10 CER, Discussion 10B (Is the Land Ethic Science, Ethics, or Both?), and your Quarter 4 capstone VR project. The week itself is Rachel Carson's Silent Spring, ecosystem services, and the year's one long argument from atoms to ecosystems. In the CER and the discussion, reason from the evidence and be fair to the other side — that is responsibility. Quiz due [date]. You have done real scientific thinking this year."

WEEK 36 · KIT WEEK 21 · FINAL

"No new content this week — just the review lesson tracing the whole year's argument, the review activity (please do it before the exam: it is the best study you can do), and the Final Exam ([exam date/time]): 70 multiple-choice questions and 3 short CER responses, covering Modules 1 through 10. Revisit any topic where your quiz confidence was low. The final asks you to see the connections, not just the facts — show what you can reason."

Situational scripts

MID-MODULE CHECK-IN

"If [topic] felt shaky on the last quiz, you're not alone — it is one of the most commonly missed ideas at this point in the year. [One-sentence re-explanation or pointer back to the relevant lesson]. Come find me at [contact] if you want to talk it through."

PRE-MAJOR-ASSESSMENT

"[Assessment name] is due [date]. The strongest submissions so far have all done one thing: [specific tip — e.g., named the specific example instead of gesturing at it]. You have what you need — go show your reasoning."

DISCUSSION NUDGE

"Discussion [N] closes [date]. A strong post names its evidence specifically and says why it supports the claim; a strong reply responds to a particular point a classmate made. Post early in the week so there is time to reply."

RE-ENGAGEMENT NUDGE (QUIET STUDENTS)

"Hi [student name] — I noticed you haven't submitted [item] yet. No judgment; just checking in. Is something in the way, or do you need more time? Reply here or reach me at [contact] and we'll figure it out together."

SECTION 10 · TEACHER WORKFLOW & WORKLOAD MAP

The 500:1 arithmetic

Optima's Canvas design sets three kinds of graded learning moment in a deliberate ratio — **about 60% Student-and-Material** (auto-graded, no teacher cost), **about 25% Student-and-Community** (discussions, light teacher presence), and **about 15% Student-and-Teacher** (the irreplaceable deep-feedback moment) — and budgets about 5 minutes per Tier 3 submission and about 30 seconds per discussion post. Here is what the real Semester 2 counts produce.

Where the graded items fall (counts from the kit)

Layer	Items	Count	Share of 155 graded items
Student & Material (auto or completion/minimal review)	80 paired assignments + 20 reading quizzes + 40 VR + 1 review activity	141	91%
Student & Community	6 discussions	6	4%
Student & Teacher (rubric / hand-scored)	5 module CERs + 2 quarterly projects + the Final's 3 CER responses (the Final counted once)	8	5%

Honest reading: by item count the kit is much more Material-heavy (about 91%) and much lighter on Community (about 4%) than the 60/25/15 design target — the six discussions are the only community items. By *grade weight* the picture is different: Discussion & Participation is 15% of the grade and CER Writing is 30%. If you want a livelier community layer, the discussion windows are the place to invest your own commentary rather than adding assignments.

Teacher-hours for a 500-student roster (estimates)

Block	Arithmetic (estimate)	Hours
One module CER	$500 \text{ students} \times 5 \text{ min} = 2,500 \text{ min}$	41.7
One discussion (initial post + reply)	$500 \times 2 \text{ posts} \times 30 \text{ s} = 500 \text{ min}$; plus ~10 min of public comments	8.5
One quarterly VR project (assumed 3 min each)	$500 \times 3 \text{ min} = 1,500 \text{ min}$ — <i>my assumption; no source states a time</i>	25.0
Final: 3 CER responses	$500 \times 3 \text{ items} \times 2 \text{ min} = 3,000 \text{ min}$ (from the Final specification)	50.0
Final Review Activity (assumed 1.5 min each)	$500 \times 1.5 \text{ min} = 750 \text{ min}$ — <i>assumption</i>	12.5

Module	What lands	Est. hours
Module 6	CER + D6	50
Module 7	CER + D7 + Q3 project	75
Module 8	CER + D8	50
Module 9	CER + D9 + Q4 project	75
Module 10	CER + D10A + D10B	59
Week 36	Final CER x3 + review activity	62
Semester total	5 CER + 6 discussions + 2 projects + Final	372

That is roughly 372 hours across 21 weeks — an average of about 18 hours a week — for 500 students, or about 19 hours for a section of 25 (the same per-student minutes, scaled). All figures are planning estimates, not measured. Paired assignments, quizzes and VR assignments are budgeted at about 0 hours (auto, completion or spot-check); the weekly quiz-statistics skim is a few minutes.

A tension to plan around. The Optima design envelope for a 500-student teacher is about 8–12 hours of Tier 3 work per module cycle — yet its own arithmetic ($500 \times 5 \text{ min} \approx 41\text{--}42$ hours per assessment, as above) shows a full 5-minute read of every module CER alone exceeds that envelope. If your roster is large, use the course's own levers: shorter 2–3 minute checkpoint reads on some CERs, Ignite AI draft feedback you edit (Section 4), batch grading in one sitting per module, and the rubric's fixed criteria so scoring stays fast. Do not cut the contested-question CERs (M9, M10) short; those are where your judgment matters most.

A normal week's workflow

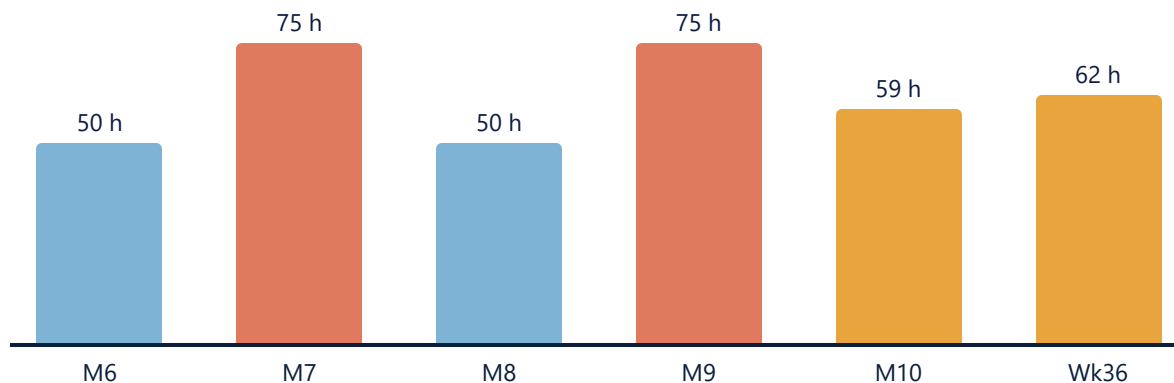
1) Skim last week's reading-quiz item analysis for a misconception cluster. 2) Spot-check a sample of paired assignments and VR captures. 3) If a CER, discussion, or project is due, grade it in one batched sitting using the Section 6 rubrics. 4) Post the weekly announcement (Section 9) and, in discussion weeks, your two or three public comments. 5) Answer student messages, using Ignite AI to draft where helpful.

Heavy weeks to flag on your own calendar

- **Week 22 (M7)** — the Module 7 CER and D7 are both due; the Q3 teacher-feedback checkpoint also falls in Weeks 21–22.
- **Week 25 (M8)** — the Module 8 CER and D8 are due.
- **Weeks 26–27** — the Module 9 CER opens, the Q3 project is due at the end of Week 27, and the Q3 gallery walk is assembled.
- **Week 29 (M9)** — the Module 9 CER and D9 are due; the vaccine and reproduction lessons carry the semester's family-notice content (Section 13).
- **Week 35 (M10)** — the heaviest week of the semester: the Module 10 CER, the Q4 capstone project, D10B, and the Week 35 quiz all land together, the week before the Final.
- **Week 36** — hand-scoring 3 CER responses per student, plus checking the review activity, on a tight turnaround before grades are due.

Not yet done. The kit has not been imported into Canvas, so none of the entering/QA tasks that follow an import (confirming group weights, rubric attachment, discussion settings, the Final's item count, due dates) have been performed. Budget a first-import verification pass before Week 16 goes live.

Workload-fluctuation graphic



Estimated teacher-grading hours by module for a 500-student roster, from the arithmetic above — estimates, not measured minutes. Blue = lighter, gold = moderate, coral = heaviest. The bars total each block's CER, discussion, project and Final hours; they do not show the week-by-week concentration, which is why the heavy-weeks list matters (Week 35 especially).

SECTION 11 · STANDARDS COVERAGE MAP (TEACHER-ONLY — NEVER STUDENT-FACING)

Status of these codes. The codes below are the Florida standards tracked in the course design (CLAUDE .md, the module table). For Semester 1 the code list was cross-checked against the official CPALMS Biology 1 (#2000310) standards list, which found several codes that did not match. **That cross-check has not been done for the Semester 2 codes [TO BE CONFIRMED]**, so this table gives codes, where they are taught and where they are assessed — and deliberately does *not* reproduce benchmark wording that I could not source. Two codes appear under two modules in the course design: L.14.7 (Modules 7 and 8) and L.16.10 (Modules 5 and 9).

By module (quick reference)

Module	Standards carried (tracked in the course design)	Taught	Assessed by
M6 · Evolution and Natural Selection	L.15.1, L.15.2, L.15.3, L.15.13, L.15.14, L.15.15	Wks 16–18	Quizzes Wks 16–18 (27 items); Module 6 CER; D6; Final: 8 MC
M7 · Classification and Diversity of Life	L.15.4, L.15.5, L.15.6, L.15.8, L.14.7	Wks 19–22	Quizzes Wks 19–22 (40); Module 7 CER; D7; Q3 project; Final: 7 MC
M8 · Plant Biology	L.14.7, L.18.10, L.18.11	Wks 23–25	Quizzes Wks 23–25 (27); Module 8 CER; D8; Q3 project (plant station); Final: 5 MC

Module	Standards carried (tracked in the course design)	Taught	Assessed by
M9 · Human Body Systems	L.14.26, L.14.36, L.14.52, L.16.10	Wks 26–29	Quizzes Wks 26–29 (40); Module 9 CER; D9; Q4 project (homeostasis station); Final: 10 MC
M10 · Ecology and Environmental Science	L.17.1, L.17.2, L.17.4, L.17.5, L.17.8, L.17.9, L.17.11, L.17.16, L.17.17, L.17.19, L.17.20, E.7.1, E.7.3	Wks 30–35	Quizzes Wks 30–35 (54); Module 10 CER; D10A, D10B; Q4 project; Final: 12 MC + CER 3
Week 36 · Final	All modules cumulative	Wk 36 review	Final: 70 MC + 3 CER; review activity

Code ranges are expanded from the course design's shorthand (for example L.15.1–15.3 is shown as L.15.1, L.15.2, L.15.3) without checking that every intermediate code exists in the official list. All codes carry the Florida prefix SC.912. They never appear on any student-facing page, per the course's hard rules; this table is one of the few places they exist.

Topic-to-week crosswalk (what is actually taught where)

Module	Topics, by week
M6	Wk 16 fossil record, comparative anatomy, biogeography, molecular evidence · Wk 17 natural selection, selection types and artificial selection, drift/gene flow/variation, antibiotic resistance · Wk 18 speciation, Hardy–Weinberg, origin of life, human evolution
M7	Wk 19 Linnaean hierarchy, binomial names, keys (use, construct) · Wk 20 cladograms, monophyletic groups, molecular phylogenetics, revisable classification · Wk 21 three domains, Bacteria/Archaea, protists, fungi · Wk 22 plants, animals, viruses, Florida biodiversity
M8	Wk 23 plant cell, tissue systems, roots/stems, leaf · Wk 24 xylem/cohesion-tension, transpiration/stomata/potometer, phloem, soil and microbiome · Wk 25 flowers, double fertilization/seed/fruit/germination, hormones, tropisms
M9	Wk 26 organization, homeostasis, skin, skeletal/muscular · Wk 27 digestion, circulation, cardiovascular health, respiration · Wk 28 kidney, neuron, synapses/drug mechanism, endocrine · Wk 29 innate and adaptive immunity, vaccines, reproduction/integration
M10	Wk 30 levels, interactions, Florida ecosystems, biomes · Wk 31 food webs, 10% rule, biomagnification/keystone, Yellowstone · Wk 32 carbon, nitrogen, water/phosphorus cycles, disruption · Wk 33 growth models, carrying capacity, succession, climate evidence · Wk 34 biodiversity, loss, invasives, conservation · Wk 35 Silent Spring, services, synthesis, humans as members and shapers

Use this crosswalk with the official standards list to complete a code-by-code alignment document if one is needed externally; raise the unconfirmed codes with the course owner first.

SECTION 12 · WHAT MAKES THIS COURSE CLASSICAL & CULTIVATING CULTURE

What makes it classical

The microscopic-to-macroscopic spine, completed. Semester 1 ran from the nature of science to the chemistry of life, the cell, energy and heredity. Semester 2 carries the one long argument outward: heritable variation (M5) becomes the raw material of natural selection and common descent (M6); common descent produces the nested pattern that classification records (M7); plants are the photosynthetic foundation (M8); the human body is the integrated organism (M9); and every organism lives in webs of energy and matter (M10). Lessons name the connection explicitly — for example the Na^+/K^+ pump from the cell module returns in the neuron (Lesson 28.02), and the carbon atom of Module 4 returns in the carbon cycle (Lesson 32.01) — and the Week 36 review lesson and activity make the bridges themselves the thing students are assessed on.

CER as classical rhetoric. Claim-Evidence-Reasoning maps onto the classical argument (claim as thesis, evidence as proof, reasoning as logos). The five Semester 2 CERs scale the same disciplined argument from guided open-inquiry (M6–M8) to contested questions (M9, M10), and the Final compresses it into three timed paragraphs.

History-of-science figures, module by module. M6: Darwin and Wallace, Pasteur. M7: Linnaeus, Aristotle, Woese, Leeuwenhoek the draper. M8: Malpighi and Grew, Stephen Hales, Darwin and his son. M9: Bernard and Cannon (homeostasis), Harvey (vs. 1,400 years of Galen), Banting and Best, Jenner. M10: Humboldt, Elton, Priestley/Lavoisier/Ingenhousz, Hellriegel and Wilfarth, Leopold, Carson. These are not decoration: they model how scientific understanding forms, stalls, and is revised.

Virtue threads made concrete. M6 Courage (Darwin following evidence into twenty years of controversy — and the courage to commit to a claim in D6); M7 Courtesy (the ordered naming of the household of life — and the courteous

disagreement of D7); M8 Wonder (a tree lifting water with no pump; Darwin's patient watching); M9 Self-government (homeostasis as the body's quiet discipline, and addiction as that machinery captured); M10 Responsibility (Leopold's land ethic and Carson's *Silent Spring*; stewardship and the common good).

Cultivating culture

- Name good thinking specifically in feedback and announcements — not “good job,” but “you separated what the wolves clearly did from what is still debated.”
- Let your weekly announcement model curiosity out loud (Section 9): reference the virtue thread, ask a genuine question, do not just recite due dates.
- Use the six discussions as the place a real intellectual community forms — substantive posts and replies, not compliance posts — and use your public comments to model the move you want (restate before you respond; define your terms; separate is from ought).
- Keep the virtue thread visible every module, not only at the CER. It is a lens for how you talk about the content, not a separate assignment.
- In an asynchronous section, a specific, warm note on a CER does more for a student's sense of being known than any policy can.

SECTION 13 · SENSITIVE-CONTENT FLAGS & FAMILY/STUDENT SCRIPTS

A systematic keyword search of the Module 6–10 and Week 36 build files (terms including contraception, abortion, puberty, sexually transmitted, intercourse, pregnancy, overdose, suicide) returned no matches, so the flags below are the contested or sensitive topics identified by reading the lessons. These are advance-notice and framing suggestions. The sample notices are language to adapt, not Optima policy; confirm tone and any required procedures with school leadership **[TO BE CONFIRMED]**. Keep every conversation respectful, factual and non-political, and keep drug content at the level of mechanism.

Flag 1 — Module 6: evolution, human evolution, and the origin of life (Weeks 16–18)

Module 6 is the course's sustained treatment of evolution, with Lesson 18.04 (“Our Own Branch: Human Evolution”) presenting the hominid lineage as a branching

bush and correcting “humans evolved from chimpanzees” (humans and chimpanzees share a common ancestor roughly 6–7 million years ago), and the Week 16 VR-B walk of a hominid skull sequence. Lesson 18.03 handles the origin of life with care: Miller–Urey is read precisely (building blocks form; no cell is created), the RNA-world hypothesis is labeled a hypothesis, and the lesson states that how the first cell arose is unsolved. It also distinguishes the disproved idea of spontaneous generation from that open question. Students are never asked to adopt or give up a belief; they are asked to understand and explain the evidence.

SUGGESTED ADVANCE NOTICE · SEND BEFORE WEEK 16

“This semester begins with Module 6, Evolution and Natural Selection, the central organizing theory of modern biology. Students will examine four independent lines of evidence (fossils, anatomy, geography, and DNA), the mechanisms of change, and, in Week 18, the human lineage and an honest look at how little is yet known about the origin of life. We teach it as the scientific community understands it, with the evidence in view, and invite students to reason from it using the claim-evidence-reasoning skills they have practiced all year. If you would like to talk with your student about this module, or have questions about how it is presented, please contact me at [contact].”

Flag 2 — Module 9, Week 28: drug mechanisms and addiction (Lesson 28.03; Module 9 CER; Week 28 VR-B)

The lesson, VR-B and CER trace how nicotine, caffeine, cocaine/amphetamines and opioids act at the synapse (agonist, antagonist, reuptake blocker), why many converge on the dopamine reward pathway, and how neuroadaptation produces tolerance and dependence. The lesson frames it explicitly as “biology, never as instructions,” and the VR-B note says to keep the framing clinical and mechanistic. The CER then asks for a reasoned position on self-government and responsibility, with the counterargument that genetics and environment matter beyond the synapse. Teacher moves: keep every conversation at the level of mechanism; do not discuss sourcing, dosing or use; treat any other drug a student proposes for the CER as a mechanism question only (approve it only if the student can describe its synaptic action correctly); remind students not to share personal medical information in discussion (the D9 teacher note). If a student discloses something about themselves or their family, follow your school’s counselor and reporting procedures **[TO BE CONFIRMED]** rather than handling it in the discussion thread.

SUGGESTED ADVANCE NOTICE · SEND BEFORE WEEK 28

"Next week in Biology 1 we study how nerve cells communicate across synapses, and, as part of that, how common substances such as caffeine, nicotine, and certain medications and drugs change that signal. The treatment is strictly scientific and mechanistic: how a molecule fits a receptor, what that does to a signal, and why repeated use can lead to tolerance and dependence. Nothing in the lesson describes how to obtain or use any substance. Students also reflect on what understanding this biology means for self-government. If you would like to talk with your student first, please reach out at [contact]."

Flag 3 — Module 9, Week 29: the reproductive system (Lesson 29.04)

Lesson 29.04 describes the reproductive system at the level of gametes, fertilization, hormonal coordination, and the link back to meiosis from Module 5, then closes the module with a body-integration activity. The content read is anatomical and mechanistic; no discussion of sexual health, contraception or pregnancy outcomes was found by keyword search. Some families still prefer advance notice of any reproductive-system content.

SUGGESTED ADVANCE NOTICE · SEND BEFORE WEEK 29

"This week Biology 1 completes the human body systems with the immune system and, in the final lesson, the reproductive system, covered at the level of gametes, fertilization, hormones, and the link to meiosis, as one body system among many. If you would like to preview the lesson or talk with your student about it, please contact me at [contact]."

Flag 4 — Module 9, Week 29: vaccines (Lessons 29.02–29.03)

Lesson 29.03 explains how vaccines work from Jenner to mRNA, why a vaccine cannot cause the disease it prevents (an antigen, not an active pathogen), and herd immunity, with an in-lesson CER practice. The mRNA callout is hedged ("much of this work is still early and unproven"). Teacher moves: teach the immunology as biology; do not give medical advice or discuss individual families' vaccination decisions; if a student raises a personal or family view, acknowledge it and bring the conversation back to the mechanism and the evidence in the lesson. No separate notice is strictly required, but this is a topic where some families value one.

"This week students study how the immune system works, including how vaccines train it. The lesson covers the biology only (antigens, memory cells, and why a vaccine cannot cause the disease it prevents), not medical advice about any individual's health decisions, which belong with you and your doctor. Questions are welcome at [contact]."

Flag 5 — Module 10: climate evidence and stewardship (Lesson 33.04; Module 10 CER; D10B; Weeks 34–35)

Lesson 33.04 and the Module 10 CER ask students to evaluate three independent lines of evidence for human-driven climate change and then reason toward a stewardship position; D10B asks whether Leopold's land ethic is science, ethics or both; Week 35 covers *Silent Spring* and the policy change it prompted. The course's own framing is deliberate and worth repeating in your own words: the lesson says the evidence is treated "soberly, from evidence, without alarmism and without dismissal"; it separates a strong consensus on the evidence from open uncertainty about rate and magnitude; and the CER names the genuinely open question as what to do about it — a question of values and stewardship. Teacher moves: grade the reasoning from evidence, not the stance on policy; keep D10B on the course's evidence and off any party or policy (the prompt tells students this; step in with a public comment if it drifts); never reduce a position to a slogan in either direction.

SUGGESTED ADVANCE NOTICE · SEND BEFORE WEEK 33

"In Weeks 33–35 Biology 1 examines the evidence for a changing climate and what ecologists mean by stewardship. Students learn how three independent kinds of evidence (the temperature record, atmospheric carbon, and shifts in living things) are weighed, and we are careful to separate what the evidence shows from what individuals and communities decide to do about it, which students are asked to reason through fairly rather than adopt from us. If you have questions about how the unit is presented, please contact me at [contact]."

Other items checked

Week 34 covers Florida's invasive species (pythons, lionfish, Brazilian pepper) and biodiversity loss in a before-and-after, mechanism-oriented way; Lesson 22.03 treats whether viruses are alive as a real scientific disagreement; Lesson 27.03 (cardiovascular health) and the cardiovascular and immune content mention disease and mortality in a public-health frame. None of these seemed to need a separate family notice. No other graphic, violent or mature content was found in a

direct read of the topics above; this is a topic-level review, not a line-by-line read of all 160 reading and lesson pages **[TO BE CONFIRMED]**.

APPENDIX · OPEN ITEMS REGISTER

Every item below was marked **[TO BE CONFIRMED]** or stated as a limit in this manual, gathered in one place. None was invented to fill a gap.

#	Open item	Who decides / what resolves it
1	The kit has not been imported into a Canvas course; all “what Canvas will show” statements are the specification’s intent.	Run a test import; verify groups, weights, rubrics, discussions, Final (70 MC + 3 essays, 88 pts, 1 attempt).
2	No Canvas course exists for Semester 2 ; it arrives as a Course Kit.	Informational; import per the Course Kit workflow.
3	The 28 Semester 1 items in the Final are not separately QA’d in this build.	Teacher or course owner reads the M1–M5 items in <code>final-quiz.md</code> .
4	The kit carries no due dates ; intended due weeks only.	Teacher sets calendar dates after import.
5	Module intro/outro wrapper quizzes are not in the kit.	Course owner: build or decline.
6	Semester 2 standards codes are not cross-checked against the official CPALMS list; two codes appear under two modules.	Course owner, against the CPALMS export.
7	Ignite AI feature set and availability.	Check your Canvas admin settings.
8	The Final has no specified time limit ; Final CER items have no attached Canvas rubric.	Teacher sets; build a Canvas rubric from Section 6 if wanted.
9	Whether the teacher manual PDF is bundled in Teacher Resources.	Per the specification, when present; confirm after the kit is rebuilt.
10	How the Semester 1 and 2 gradebooks combine into a year grade , given that the Exams group holds only the Final here.	School decision.
11	Workload minutes for the Q3/Q4 projects (3 min) and the review activity (1.5 min) are my assumptions.	Test in your first cycle and adjust.

#	Open item	Who decides / what resolves it
12	The note block on the 40 VR pages and 2 project pages is student-facing and its heading uses an older term for the teacher.	Course owner: relabel or hide.
13	The family-notice language , counselor/reporting procedures, and any drug-disclosure protocol in Section 13.	School leadership.
14	The sensitive-content review is a topic-level read, not a line-by-line read of every page.	Teacher spot-reads Weeks 18, 28, 29, 33 before teaching.

Optima Academy Online · Biology 1 · Teacher Manual (Semester 2 Edition)

Teacher copy. Contains answer keys and grading guidance. Do not publish to students.